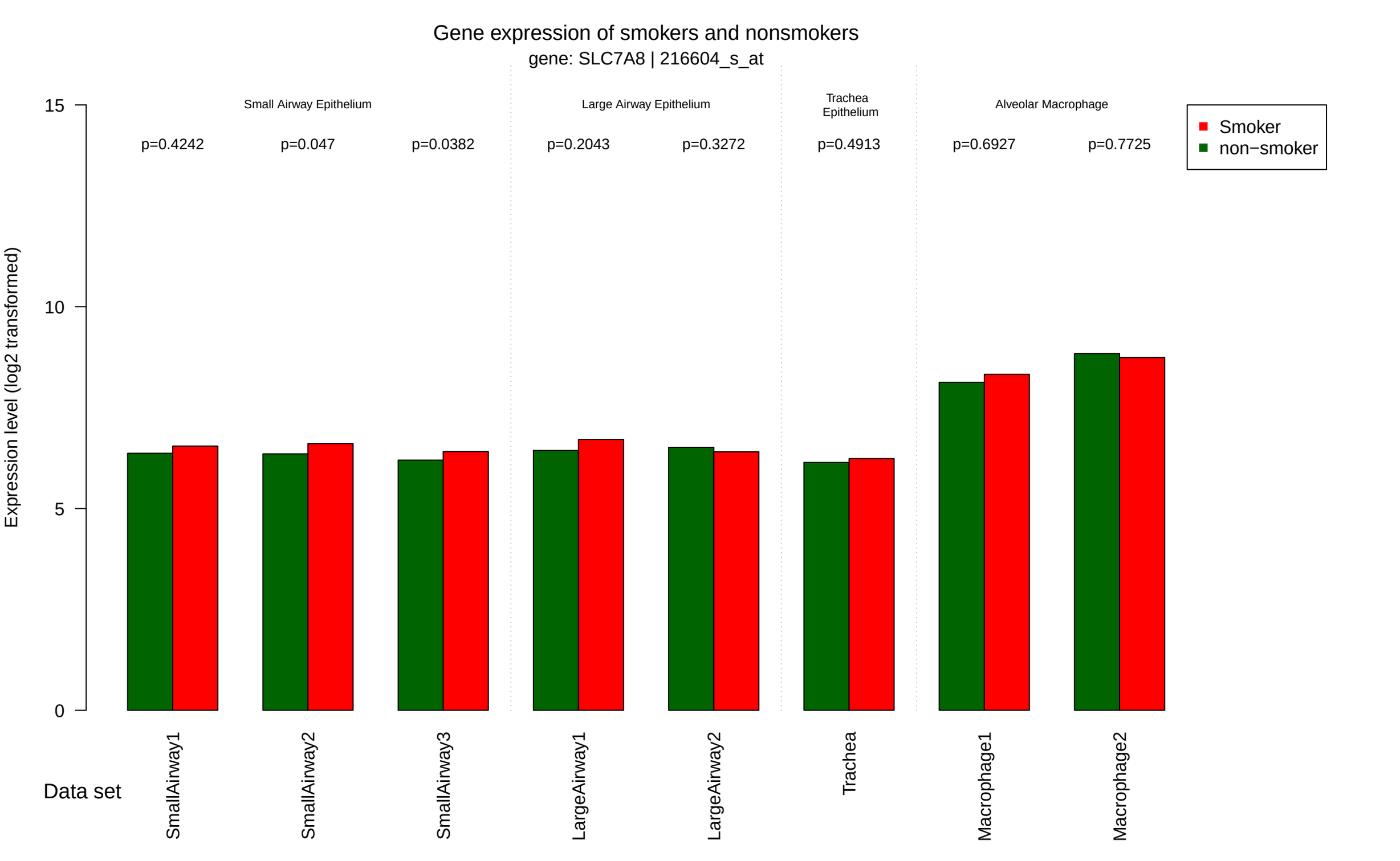
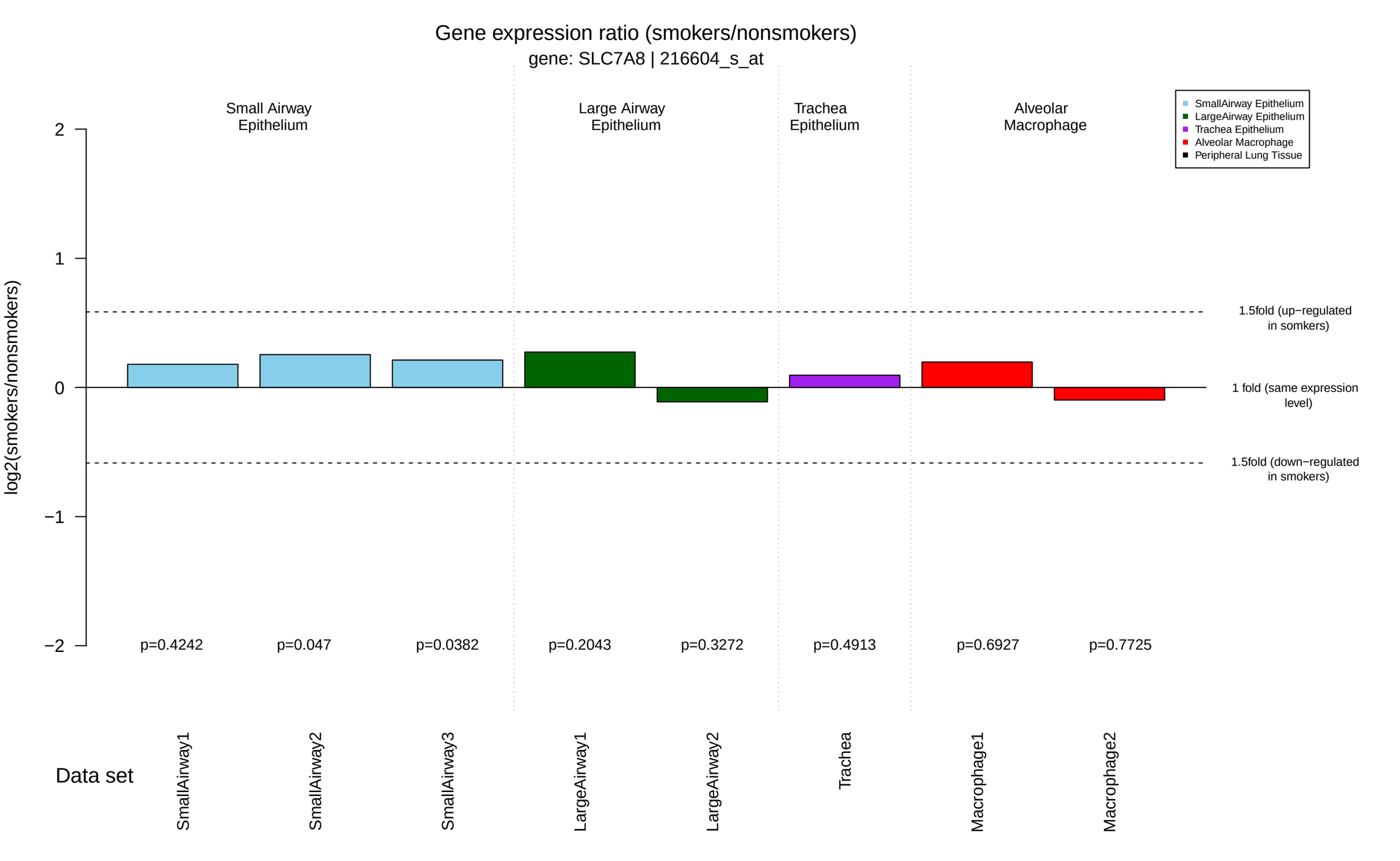
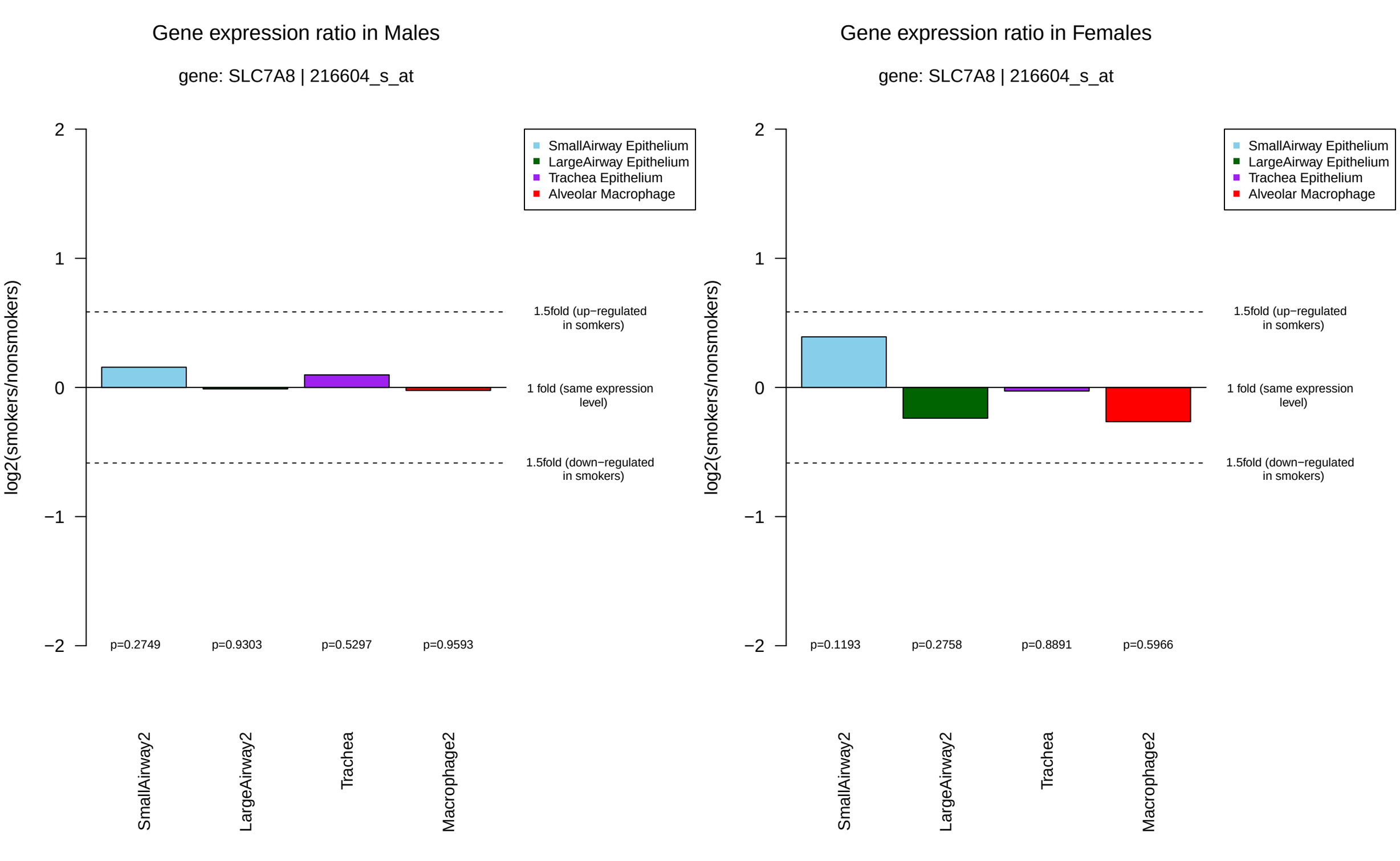
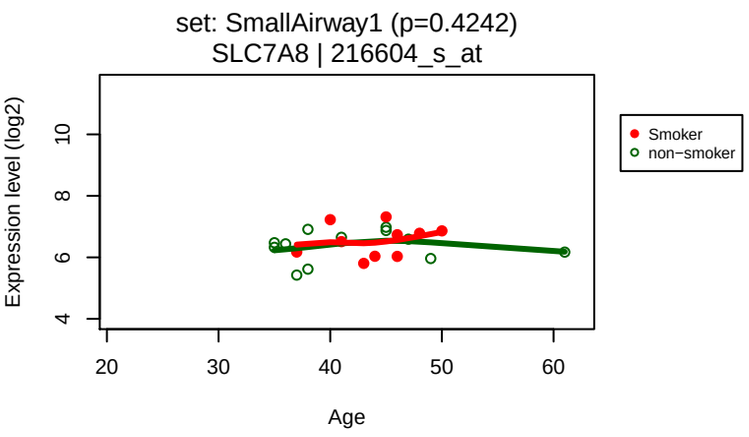
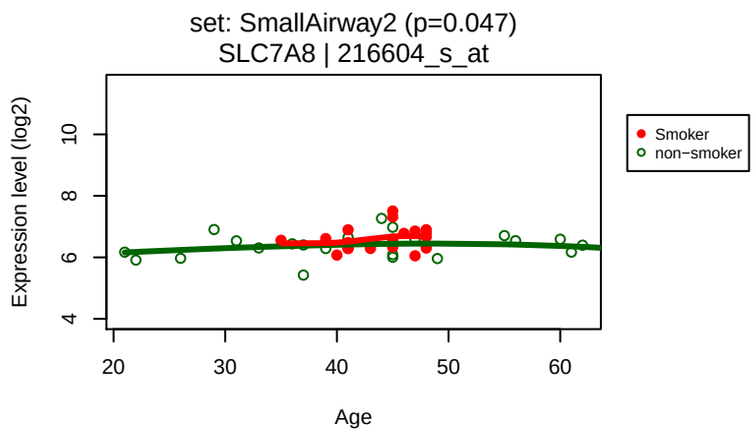
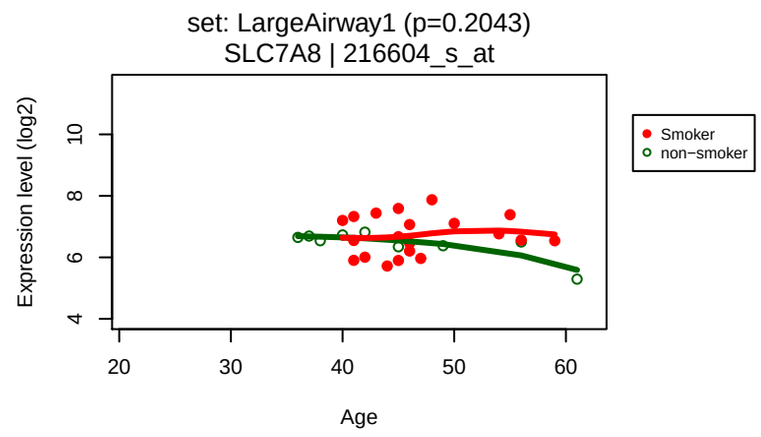

Gene expression of smokers and nonsmokers
gene: SLC7A8 | 216604_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4242
p=0.047
p=0.0382
p=0.2043
p=0.3272
p=0.4913
p=0.6927
p=0.7725
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

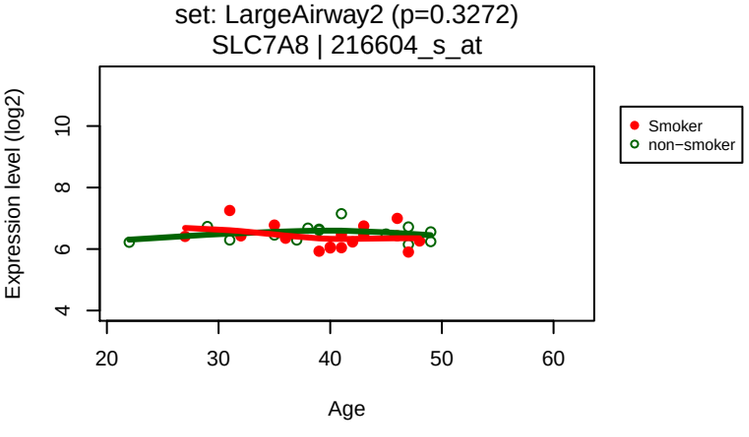

Gene expression ratio (smokers/nonsmokers)
gene: SLC7A8 | 216604_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4242
p=0.047
p=0.0382
p=0.2043
p=0.3272
p=0.4913
p=0.6927
p=0.7725
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

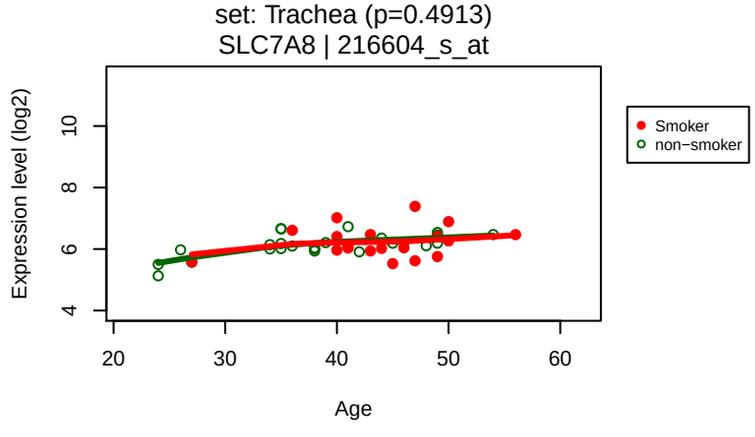

Gene expression ratio in Males
gene: SLC7A8 | 216604_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2749
p=0.9303
p=0.5297
p=0.9593
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC7A8 | 216604_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1193
p=0.2758
p=0.8891
p=0.5966
SmallAirway2
LargeAirway2
Trachea
Macrophage2

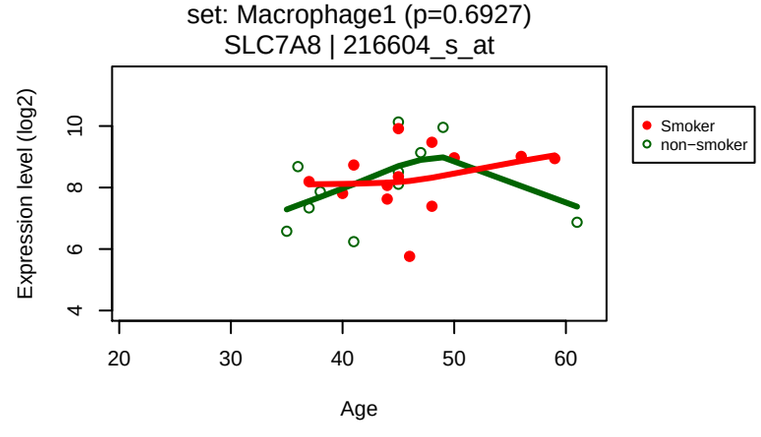

set: SmallAirway1 (p=0.4242)
SLC7A8 | 216604_s_at
Expression level (log2)
Age
Smoker
non−smoker

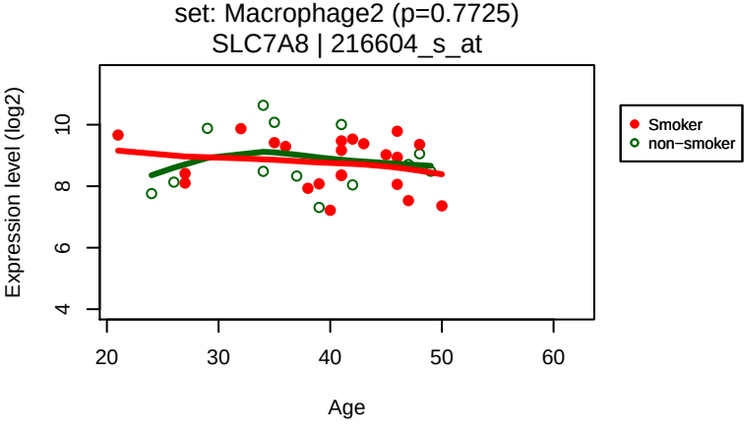

set: SmallAirway2 (p=0.047)
SLC7A8 | 216604_s_at
Expression level (log2)
Age
Smoker
non−smoker

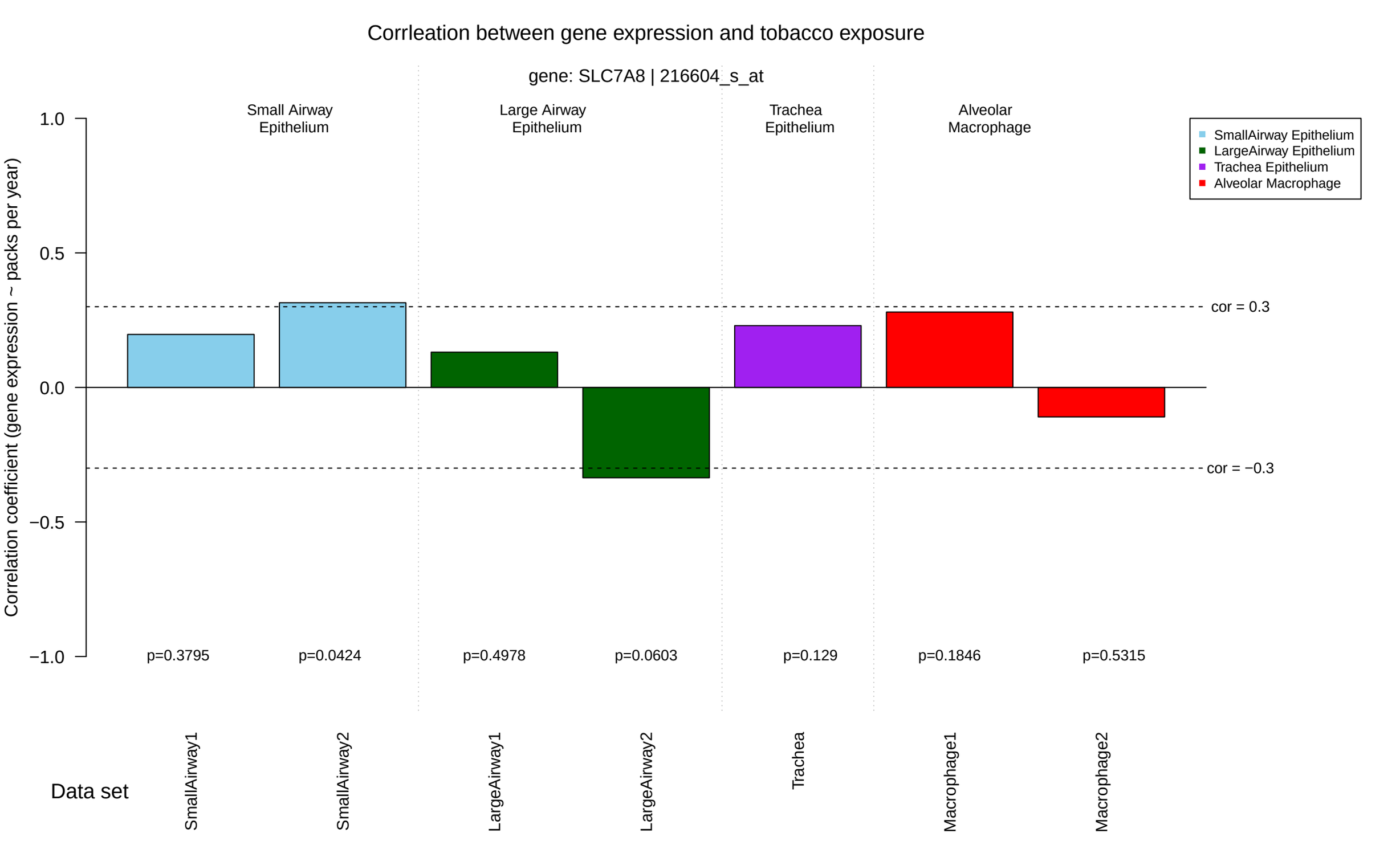

set: LargeAirway1 (p=0.2043)
SLC7A8 | 216604_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3272)
SLC7A8 | 216604_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4913)
SLC7A8 | 216604_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6927)
SLC7A8 | 216604_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7725)
SLC7A8 | 216604_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC7A8 | 216604_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3795
p=0.0424
p=0.4978
p=0.0603
p=0.129
p=0.1846
p=0.5315
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2